AS · Unit 1 · WPH11 Mechanics & Materials

Stretching Materials

Hang a mass on a spring and it stretches; double the mass and it stretches twice as far. Pull a copper wire hard enough and it stays longer when you let go; pull a glass rod and it simply snaps. This topic gives you the language and the graphs to describe all of that: **Hooke's law** and **stiffness** for objects, **stress**, **strain** and the **Young modulus** for materials, and the **elastic strain energy** stored when something is stretched.

Hooke's law and stiffness

Hang a 100 g mass on a spring and it stretches by, say, 4 cm. Add another 100 g and it stretches another 4 cm. Every equal step in force gives an equal step in **extension**, so the extension is directly proportional to the force. That is Hooke's law.

Hooke's law

FORMULA SHEET

$$F = k\,\Delta x$$

F: force applied / N k: stiffness / N m⁻¹ Δx: extension (or compression) / m

The constant k is the **stiffness** of the object: the force needed per metre of extension. A stiff spring has a large k and barely stretches; a floppy one has a small k. On a graph of force (y-axis) against extension (x-axis), Hooke's law gives a straight line through the origin, and its **gradient is** k.

DEFINITION

Hooke's law: the extension of an object is directly proportional to the force applied to it, provided the limit of proportionality is not exceeded.

Stiffness k: the force per unit extension, $k = F/\Delta x$, measured in N m⁻¹.

Hooke's law works for squashing too. Push on the ends of a spring and its **compression** Δx (the decrease in length) is proportional to the force, with the same equation. Stiffness belongs to the **object**: a thicker wire, a shorter spring or a spring made of stiffer metal all have a bigger k. We'll fix that with the Young modulus in a moment.

COMMON MISTAKE

Extension is the **change** in length, not the new length. A spring with a natural length of 12.0 cm that is stretched to 15.0 cm has $\Delta x = 3.0\ \text{cm} = 0.030\ \text{m}$. And always convert cm or mm to metres before using $F = k\Delta x$.

WORKED EXAMPLE

A spring has a natural length of 80 mm. When a 300 g mass hangs from it, its length is 104 mm. (a) Calculate the stiffness of the spring. (b) Calculate the length of the spring when the load is increased to 450 g, assuming the limit of proportionality is not exceeded.

Show solution

(a) The force is the weight of the mass, and the extension is the change in length:

$$F = mg = 0.300 \times 9.81 = 2.94\ \text{N}, \qquad \Delta x = 104 - 80 = 24\ \text{mm} = 0.024\ \text{m}$$

$$k = \frac{F}{\Delta x} = \frac{2.94}{0.024} = 123\ \text{N m}^{-1}$$

(b) Extension is proportional to the load, so 450 g gives $\frac{450}{300} \times 24 = 36\ \text{mm}$ of extension. The length is $80 + 36 = 116\ \text{mm}$.

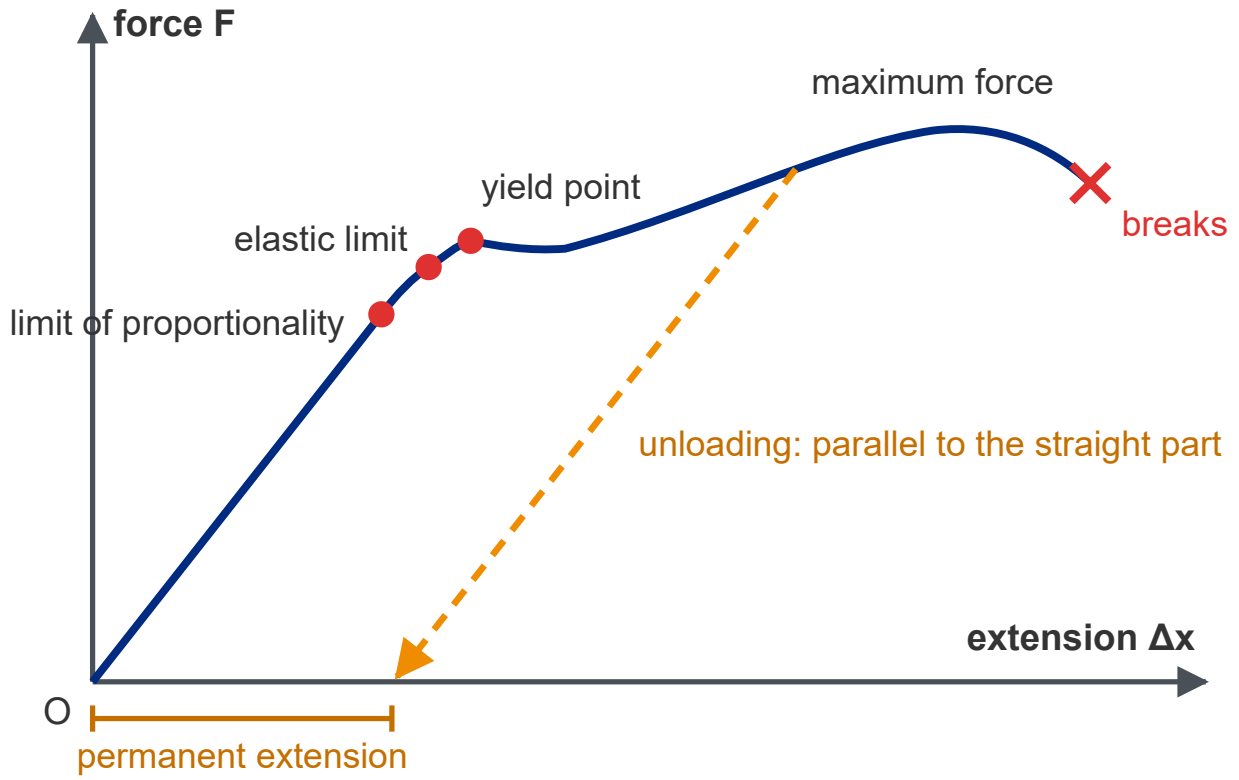
Using ratios like this avoids rounding errors, and it doesn't even need g.

EXAM TIP

Check the axes before you read a gradient. Edexcel usually puts force on the y-axis, so the gradient is k. If a graph has extension on the y-axis and force on the x-axis, the gradient is $1/k$, and a steeper line means a *less* stiff spring.

Force–extension graphs: elastic and plastic

Load a metal wire until it breaks and the force–extension graph tells the whole story. It starts as a straight line (Hooke's law), then bends over, and eventually the wire starts to stretch a lot for very little extra force before it finally breaks.



A metal wire stretched to breaking (not to scale). Up to the elastic limit the deformation is elastic. Beyond it the deformation is plastic: unloading follows a line parallel to the straight part and leaves a permanent extension.

DEFINITION

Limit of proportionality: the point beyond which force is no longer proportional to extension (Hooke's law stops applying).

Elastic limit: the maximum force (or extension) for which the object will still return to its original length when the force is removed.

Yield point: the point beyond which the material stretches a lot (plastically) with little or no increase in force.

Elastic deformation: the object returns to its original shape and size when the deforming force is removed.

Plastic deformation: the object does not return to its original shape when the force is removed; it is permanently deformed.

COMMON MISTAKE

The limit of proportionality and the elastic limit are **not** the same thing. Between them the graph is already curved, but the wire still springs back to its original length. Rubber is the extreme case: it is elastic up to huge extensions but its force–extension graph is curved almost everywhere.

Why the unloading line is parallel

Force–compression graphs

For a compressive force you plot force against **compression** and the same ideas apply: a straight section where Hooke's law holds (gradient = stiffness), and elastic or plastic behaviour beyond it. Squash a spring and the graph is straight until the coils touch; after that it shoots up almost vertically, because you are now compressing solid metal. Squash a lump of modelling clay and it deforms plastically almost straight away.

Stress, strain and the Young modulus

Stiffness describes one particular spring or wire. Engineers want a number that describes the **material**, whatever its shape. Two thought experiments show how to get one:

- Hang a load from **two identical wires side by side**. Each carries half the load, so each stretches half as much. Doubling the cross-sectional area halves the extension, so what matters is the **force per unit area**.
- Use a wire **twice as long**. Each half carries the full load and stretches just as much as a single short wire, so the total extension doubles. What matters is the **extension per unit length**.

Stress

FORMULA SHEET

$$\sigma = \frac{F}{A}$$

F: force / N A: cross-sectional area / m²

σ: stress / Pa

Strain

FORMULA SHEET

$$\varepsilon = \frac{\Delta x}{x}$$

Δx: extension / m x: original length / m

ε: strain (no unit)

Young modulus

FORMULA SHEET

$$E = \frac{\sigma}{\varepsilon}$$

E: Young modulus / Pa

valid in the Hooke's law region

Stress has the same unit as pressure, the pascal (1 Pa = 1 N m^{-2}); for metals you'll usually see MPa (10^6 Pa). **Strain** is a length divided by a length, so it has **no unit** (sometimes it's given as a percentage). That means the

Young modulus also has the unit Pa, usually quoted in GPa (10^9 Pa). The same definitions work for **tensile** (stretching) and **compressive** (squashing) forces.

DEFINITION

Stress: force per unit cross-sectional area, $\sigma = F/A$.

Strain: extension (or compression) per unit original length, $\varepsilon = \Delta x/x$.

Young modulus: the ratio of stress to strain, $E = \sigma/\varepsilon$, for a material obeying Hooke's law. It measures the stiffness of the material.

Linking stiffness to the Young modulus

1 Substitute the definitions of stress and strain into $E = \sigma/\varepsilon$:

$$E = \frac{F/A}{\Delta x/x} = \frac{F\,x}{A\,\Delta x}$$

This form is the one you use with raw measurements, for example in Core Practical 3.

2 Rearrange to get $F/\Delta x$, which is the stiffness:

$$\boxed{k = \frac{F}{\Delta x} = \frac{EA}{x}}$$

A thicker wire (bigger A) is stiffer, a longer wire (bigger x) is less stiff, and a stiffer material (bigger E) gives a stiffer wire, exactly as the thought experiments said.

Material	Young modulus / GPa	Material	Young modulus / GPa
rubber	0.01 to 0.1	aluminium	70
nylon	≈ 3	glass	≈ 70
bone (along its length)	≈ 17	titanium alloy	≈ 110
concrete	≈ 30	copper	≈ 120
wood (along the grain)	≈ 10	steel	≈ 200

These numbers matter in real design. A replacement hip joint ("spare-part surgery") has to be strong and must not be too much stiffer than the bone it is fixed into, which is one reason titanium alloys are used rather than steel. Food scientists test the "crunch" of a biscuit by compressing it and measuring the force, and the stiffness and breaking point tell them how it will feel to bite.

WORKED EXAMPLE

A steel wire of length 2.50 m and diameter 0.50 mm supports a load of 60 N. Its extension is 3.8 mm. Calculate (a) the stress, (b) the strain and (c) the Young modulus of the steel.

Show solution

(a) Cross-sectional area from the diameter ($r = 0.25$ mm):

$$A = \pi r^2 = \pi(0.25 \times 10^{-3})^2 = 1.96 \times 10^{-7}\ \text{m}^2$$

$$\sigma = \frac{F}{A} = \frac{60}{1.96 \times 10^{-7}} = 3.06 \times 10^{8}\ \text{Pa}$$

(b) $\varepsilon = \dfrac{\Delta x}{x} = \dfrac{3.8 \times 10^{-3}}{2.50} = 1.52 \times 10^{-3}$

(c) $E = \dfrac{\sigma}{\varepsilon} = \dfrac{3.06 \times 10^{8}}{1.52 \times 10^{-3}} = 2.0 \times 10^{11}\ \text{Pa}$ (200 GPa).

The classic slip is to use the diameter instead of the radius in πr^2, which makes the area four times too big.

WORKED EXAMPLE

When a 70 kg runner lands, the force on one thigh bone (femur) can be about three times their weight. The femur is 0.45 m long, with a cross-sectional area of 3.3 × 10⁻⁴ m² and a Young modulus of 17 GPa. Calculate the compressive stress and the amount the bone shortens.

Show solution

$$F = 3 \times 70 \times 9.81 = 2060\ \text{N}, \qquad \sigma = \frac{F}{A} = \frac{2060}{3.3 \times 10^{-4}} = 6.2 \times 10^{6}\ \text{Pa}$$

$$\varepsilon = \frac{\sigma}{E} = \frac{6.2 \times 10^{6}}{17 \times 10^{9}} = 3.7 \times 10^{-4}$$

$$\Delta x = \varepsilon x = 3.7 \times 10^{-4} \times 0.45 = 1.7 \times 10^{-4}\ \text{m} \approx 0.17\ \text{mm}$$

Tiny, and the stress is far below the breaking stress of bone (well over 100 MPa in compression), which is why bones survive running.

Stress–strain graphs and breaking stress

A stress–strain graph is a force–extension graph with the axes rescaled: divide the force by A and the extension by x. The shape is the same, but now the graph describes the **material**, not one particular sample. In the region where Hooke's law holds, the **gradient is the Young modulus**.

DEFINITION

Breaking stress: the stress at which a material breaks (fractures).

Different materials give very different curves. These words aren't spec definitions, but they are the natural way to describe what the graphs show:

- **Stiff:** a steep straight section (large E). Steel is stiffer than copper; rubber is not stiff at all.
- **Strong:** a high breaking stress.
- **Ductile:** a long plastic region, so it can be drawn out into wires. Copper is very ductile.
- **Brittle:** breaks with little or no plastic deformation. Glass obeys Hooke's law right up to the moment it snaps.
- **Rubber:** stretches to several times its length and still returns, but its graph is curved, so it never really obeys Hooke's law.

Mild steel shows a clear **yield point**: the stress suddenly stops rising (it even dips slightly) while the steel stretches a lot. After that the metal gets harder to stretch again (it "work hardens") until the maximum stress, and then it breaks. The spec only uses the term breaking stress; many books also call the maximum stress on the curve the ultimate tensile stress.

WORKED EXAMPLE

The stress–strain graph for a metal is a straight line from the origin to a stress of 240 MPa at a strain of 0.0020, after which it curves. (a) Calculate the Young modulus. (b) A wire of this metal is 2.0 m long with a cross-sectional area of 0.50 mm². Calculate the largest force for which it obeys Hooke's law, and the extension at that force.

Show solution

(a) $E = \dfrac{240 \times 10^{6}}{0.0020} = 1.2 \times 10^{11}\ \text{Pa}$

(b) Convert the area: $0.50\ \text{mm}^2 = 0.50 \times 10^{-6}\ \text{m}^2$ (1 mm² = 10⁻⁶ m²).

$$F = \sigma A = 240 \times 10^6 \times 0.50 \times 10^{-6} = 120 \text{ N}$$

$$\Delta x = \varepsilon x = 0.0020 \times 2.0 = 4.0 \times 10^{-3} \text{ m} = 4.0 \text{ mm}$$

Core Practical 3: the Young modulus of a wire

CORE PRACTICAL

Determine the Young modulus of a material

A long, thin wire (copper is typical) is clamped at one end of the bench and passes over a pulley at the other end, with a mass hanger on the end. A small marker (a piece of tape or a pointer) is fixed to the wire near the pulley, above a metre rule or vernier scale fixed to the bench.

1. Hang a small load to pull the wire straight and remove kinks. Then measure the **original length** x from the clamp to the marker with a metre rule or tape measure.
2. Measure the **diameter** with a **micrometer** at several places along the wire and in different directions; average them and calculate $A = \pi d^2/4$.
3. Add masses one at a time (for example 100 g each). After each, record the position of the marker; the extension Δx is the distance it has moved from its starting position.
4. Remove the masses one by one and check the marker returns to its start, which shows you stayed below the elastic limit.

Graph: calculate stress F/A and strain $\Delta x/x$ for each load and plot **stress against strain**. The gradient of the straight section is the Young modulus E. (Alternatively plot F against Δx: the gradient is EA/x, so $E = \text{gradient} \times x/A$.)

Why long and thin? $\Delta x = Fx/(EA)$, so a long, thin wire gives a larger extension, which can be measured with a smaller percentage uncertainty.

Sources of uncertainty and how to reduce them

- **Diameter**: the area depends on d^2, so its percentage uncertainty is doubled. This is usually the biggest uncertainty, which is why you use a micrometer and average several readings.
- **Extension**: it is small, so read the marker at eye level (avoid parallax), ideally with a vernier scale, and use a long wire.
- **Slipping** at the clamp or **friction** in the pulley give wrong extensions; make sure the clamp is tight and the pulley turns freely.
- Only use points on the straight part of the graph; points past the limit of proportionality curve away.

Safety: wear eye protection, because a wire that snaps can whip back. Put a box or sand tray under the masses and keep your feet clear.

WORKED EXAMPLE

In Core Practical 3 a copper wire has an original length of 2.50 m. Its diameter is measured as 0.38 ± 0.01 mm. A graph of load F against extension Δx is a straight line through the origin with a gradient of $5.2 \times 10^3\ \text{N m}^{-1}$. (a) Calculate the Young modulus of copper. (b) Calculate the percentage uncertainty in the cross-sectional area.

Show solution

(a) The gradient is $F/\Delta x = EA/x$, so $E = \text{gradient} \times x/A$.

$$A = \frac{\pi d^2}{4} = \frac{\pi(0.38 \times 10^{-3})^2}{4} = 1.13 \times 10^{-7}\ \text{m}^2$$

$$E = \frac{5.2 \times 10^3 \times 2.50}{1.13 \times 10^{-7}} = 1.1 \times 10^{11}\ \text{Pa}$$

(b) Percentage uncertainty in d = $0.01/0.38 \times 100 = 2.6\%$. Since $A \propto d^2$, the percentage uncertainty in A is $2 \times 2.6 = 5.3\%$. That is far bigger than the uncertainty in the 2.50 m length, which is why the diameter has to be measured so carefully.

Elastic strain energy

Stretching something takes work, and if the deformation is elastic that work is stored as **elastic strain energy**, ready to be released (think of a catapult or a bow). Work done is force × distance, but here the force isn't constant: it grows as the extension grows. The way round that is the area under the force–extension graph. Each thin strip under the graph is a small force × a small extra extension, a small piece of work, and adding all the strips gives the area.

KEY IDEA

The **area under a force–extension graph** is the work done in stretching the object. If the deformation is elastic, this is the elastic strain energy stored.

Elastic strain energy

FORMULA SHEET

$$E_{\text{el}} = \tfrac{1}{2} F \Delta x$$

E_{el}: elastic strain energy / J F: final force / N Δx: extension / m

Elastic strain energy from the area under the graph

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Below the limit of proportionality the force–extension graph is a straight line from the origin to the point $(\Delta x, F)$.

2. The area under it is a triangle with base Δx and height F:

$$E_{\text{el}} = \text{area} = \tfrac{1}{2} \times \text{base} \times \text{height}$$

$$\boxed{E_{\text{el}} = \tfrac{1}{2} F \Delta x}$$

Another way to see the ½: the force rises steadily from 0 to F, so the average force during the stretch is $F/2$, and work = average force × distance.

Elastic strain energy in terms of the stiffness

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Start from the area result:

$$E_{\text{el}} = \tfrac{1}{2} F \Delta x$$

2. Hooke's law gives $F = k\Delta x$. Substitute it in:

$$E_{\text{el}} = \tfrac{1}{2}(k\Delta x)\,\Delta x \;\Rightarrow\; \boxed{E_{\text{el}} = \tfrac{1}{2} k \Delta x^2}$$

Energy goes as the square of the extension: stretch a spring twice as far and it stores four times the energy.

COMMON MISTAKE

$\frac{1}{2}F\Delta x$ and $\frac{1}{2}k\Delta x^2$ only work while the graph is a **straight line**. For a curved graph you must find the area some other way, and if the object has been deformed **plastically**, not all the work you did is stored: the energy that went into moving atoms permanently is transferred to heating the material and is not returned when you let go.

Estimating the area under a curved graph

The spec expects you to estimate the energy for **non-linear** graphs too. The standard method is counting squares:

1. Work out what one grid square is worth: (force per square) × (extension per square) gives joules per square.
2. Count the whole squares under the curve, and count each part-square that is more than half under the curve as one (ignore those less than half).
3. Energy = number of squares × energy per square.

Alternatively split the area into simple shapes (triangles, rectangles, trapezia) and add their areas. Either way, smaller squares or thinner strips give a better estimate.

WORKED EXAMPLE

A catapult's elastic obeys Hooke's law. It is pulled back 0.15 m by a force that rises to 60 N, and it fires a 20 g stone. (a) Calculate the elastic strain energy stored. (b) Calculate the maximum possible speed of the stone. (c) Explain why the real speed will be lower.

Show solution

(a) $E_{\text{el}} = \frac{1}{2}F\Delta x = \frac{1}{2} \times 60 \times 0.15 = 4.5 \text{ J}$

(b) At most all of it becomes kinetic energy of the stone:

$$\tfrac{1}{2}mv^2 = 4.5 \;\Rightarrow\; v = \sqrt{\frac{2 \times 4.5}{0.020}} = 21 \text{ m s}^{-1}$$

(c) Some energy becomes kinetic energy of the elastic itself and some is dissipated as heat in the rubber, so the stone gets less than 4.5 J.

WORKED EXAMPLE

On a force–extension graph for a rubber band, each grid square is 0.5 N high and 1.0 cm wide. Up to an extension of 8.0 cm, a student counts 38 squares under the curve. Estimate the work done in stretching the band.

Show solution

One square represents $0.5 \text{ N} \times 0.010 \text{ m} = 5.0 \times 10^{-3} \text{ J}$.

$$W \approx 38 \times 5.0 \times 10^{-3} = 0.19 \text{ J}$$

Don't use $\frac{1}{2}F\Delta x$ here: the graph is curved, so the triangle would give the wrong area.

Quick checks

QUICK CHECK

Wires P and Q are made of the same metal. P is twice as long as Q and has twice the diameter. The same force is applied to each, within the limit of proportionality. What is (extension of P) / (extension of Q)?

- 4
- 1
- 0.5
- 0.25

$\Delta x = Fx/(AE)$. Doubling x doubles the extension, but doubling the diameter makes the area four times bigger, which divides it by 4. Overall $2/4 = 0.5$.

QUICK CHECK

Which statement describes the elastic limit?

- The point beyond which force is no longer proportional to extension.
- The largest force for which the object returns to its original length when the force is removed.
- The force at which the material breaks.
- The point where the material stretches with no increase in force.

The first option is the limit of proportionality, the third is (related to) the breaking stress and the fourth describes the yield point.

QUICK CHECK

A spring of stiffness 200 N m⁻¹ is stretched by 5.0 cm, within its limit of proportionality. How much elastic strain energy does it store?

- 0.25 J
- 0.50 J
- 5.0 J
- 250 J

$E_{\text{el}} = \frac{1}{2}k\Delta x^2 = \frac{1}{2} \times 200 \times 0.050^2 = 0.25$ J. 0.50 J forgets the ½; 250 J forgets to convert 5.0 cm into metres.

QUICK CHECK

Which of these is a correct unit for the Young modulus?

- N m⁻¹
- N m⁻²
- It has no unit, because strain has no unit.
- N m

Stress is in N m⁻² (Pa) and strain has no unit, so stress/strain is in N m⁻². N m⁻¹ is the unit of stiffness k, which belongs to an object, not a material.

QUICK CHECK

A wire is loaded beyond its elastic limit and then the load is removed. Which best describes the unloading part of its force–extension graph?

- It retraces the loading curve back to the origin.
- It is a straight line parallel to the initial straight part, ending at a positive extension.
- It drops vertically to zero force at the maximum extension.
- It is a straight line back to the origin.

The wire contracts with the same stiffness as it had in the elastic region, so the unloading line is parallel to the original straight line, but it doesn't get back to its original length: there is a permanent extension.

QUICK CHECK

A rubber band's force–extension graph curves so that it lies below the straight line joining the origin to its final point (Δx, F). The energy stored is

- exactly $\frac{1}{2}F\Delta x$
- exactly $F\Delta x$
- less than $\frac{1}{2}F\Delta x$, found from the area under the curve
- more than $\frac{1}{2}F\Delta x$, because rubber is elastic

The energy is always the area under the graph. If the curve sags below the straight line, the area under it is less than the triangle $\frac{1}{2}F\Delta x$. Estimate it by counting squares.